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1 C:\Users\Nathaniel\FarmerJackInc\CSPNuclearHybridDiss
  \.venv\Scripts\python.exe C:\Users\Nathaniel\
  FarmerJackInc\CSPNuclearHybridDiss\AP1000V5.py
2 block 0 | scalar | iterations: 52 |
  residual: 6.49e-08 | HP turbine.eta_s1
3 block 1 | scalar | iterations: 52 |
  residual: 4.97e-08 | LP turbine stage 1.eta_s1
4 block 2 | scalar | iterations: 60 |
  residual: 4.50e-08 | main condenser.subcooling
5 block 3 | scalar | iterations: 44 |
  residual: 6.92e-08 | moisture separator.
  outlet_constraint_liquid
6 block 4 | scalar | iterations: 43 |
  residual: 5.58e-08 | moisture separator.
  outlet_constraint_gas
7 block 5 | scalar | iterations: 53 |
  residual: 3.55e-08 | HP turbine.eta_s2
8 block 6 | scalar | iterations: 49 |
  residual: 4.18e-08 | LP turbine stage 1.eta_s2
9 block 7 | scalar | iterations: 59 |
  residual: 3.56e-08 | condenser pump.eta_s
10 block 8 | scalar | iterations: 53 |
  residual: 4.81e-08 | HP turbine.eta_s3
11 block 9 | scalar | iterations: 49 |
  residual: 4.90e-08 | LP turbine stage 2.eta_s
12 block 10 | scalar | iterations: 52 |
  residual: 4.38e-08 | HP turbine.eta_s4
13 block 11 | scalar | iterations: 50 |
  residual: 4.38e-08 | LP turbine stage 3.eta_s
14 block 12 | square | iterations: 60 |
  residual: 3.67e-08 | HP FWH 1.
  energy_balance_constraints, HP FWH 1.ttd_u, HP FWH 2.
  energy_balance_constraints, HP FWH 2.ttd_u, HP FWH 2
  shell merge.mass_flow_constraints, HP FWH 2 shell
  merge.energy_balance_constraints, HP FWH merge.
  mass_flow_constraints, HP FWH merge.
  energy_balance_constraints, LP FWH 1.
  energy_balance_constraints, LP FWH 2.
  energy_balance_constraints, LP FWH 2.ttd_u, LP FWH 2
  shell merge.mass_flow_constraints, LP FWH 2 shell
  merge.energy_balance_constraints, LP FWH 3.

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14 energy_balance_constraints, LP FWH 3.ttd_u, LP FWH 3
   shell merge.mass_flow_constraints, LP FWH 3 shell
   merge.energy_balance_constraints, LP FWH 4.
   energy_balance_constraints, LP FWH 4.ttd_u, LP FWH 4
   shell merge.mass_flow_constraints, LP FWH 4 shell
   merge.energy_balance_constraints, LP stage 1 exhaust
   splitter.mass_flow_constraints, LP stage 2 exhaust
   splitter.mass_flow_constraints, LP turbine stage 1.
   mass_flow_constraints, MSR drain FWH.
   energy_balance_constraints, MSR drain FWH.ttd_l,
   condenser merge.mass_flow_constraints, feed pump.
   eta_s, feedwater splitter.mass_flow_constraints,
   interstage heater 1.energy_balance_constraints,
   interstage heater 2.energy_balance_constraints,
   interstage heater drain merge.mass_flow_constraints,
   interstage heater drain merge.
   energy_balance_constraints, main steam merge.
   mass_flow_constraints, main steam merge.
   energy_balance_constraints, moisture separator.
   mass_flow_constraints, moisture separator.
   energy_balance_constraints, reheater drain FWH.
   energy_balance_constraints, reheater drain FWH.ttd_l
   , steam generator 1.Q, steam generator 2.Q
15 block 13 | scalar          | iterations: 42 |
   residual: 4.55e-08 | HP turbine.mass_flow_constraints
16 block 14 | scalar          | iterations: 65 |
   residual: 1.30e-08 | condenser merge.
   energy_balance_constraints
17 block 15 | scalar          | iterations: 44 |
   residual: 4.53e-08 | main steam splitter.
   mass_flow_constraints
18 block 16 | scalar          | iterations: 64 |
   residual: 2.19e-08 | main condenser.
   energy_balance_constraints
19 Converged: True (status=0)
20
21 ##### RESULTS (HeatExchanger) #####
22 +-----+-----+-----+-----+
   ---+-----+-----+-----+-----+-----+
   -----+-----+-----+-----+-----+-----+
   -----+-----+-----+-----+-----+-----+

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22  ----+-----+-----+
23  |               |           Q |           UA |
    kA |   td_log |   lmtd |   ttd_u |   ttd_l |
    ttd_min |   pr1 |   pr2 |   dp1 |   dp2
    |   zeta1_d4 |   zeta1 |   zeta2_d4 |   zeta2 |
    eff_hot |   eff_cold |   eff_max |
24  |-----+-----+-----+-----+
    ---+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
25  | HP FWH 1           | -5.05e+08 | 2.17e+07 | 2.17e+
    07 | 2.33e+01 | 2.33e+01 | 5.00e+00 | 6.45e+01 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 6.00e+04 | 1.89e+05
    |   1.08e+01 | 1.08e+01 |   5.80e+01 | 5.80e+01 | 7
    .80e-01 |   9.22e-01 |   9.22e-01 |
26  | HP FWH 2           | -1.57e+08 | 1.39e+07 | 1.39e+
    07 | 1.13e+01 | 1.13e+01 | 5.00e+00 | 2.16e+01 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 8.49e+04 | 1.83e+05
    |   1.47e+02 | 1.47e+02 |   5.36e+01 | 5.36e+01 | 8
    .79e-01 |   7.82e-01 |   8.79e-01 |
27  | LP FWH 1           | -1.18e+08 | 5.15e+06 | 5.15e+
    06 | 2.30e+01 | 2.30e+01 | 1.69e+01 | 3.04e+01 | 1.
    69e+01 | 9.70e-01 | 9.70e-01 | 1.80e+04 | 3.29e+04
    |   4.77e+00 | 4.77e+00 |   1.06e+01 | 1.06e+01 | 6
    .44e-01 |   4.63e-01 |   6.44e-01 |
28  | LP FWH 2           | -2.94e+08 | 1.72e+07 | 1.72e+
    07 | 1.70e+01 | 1.70e+01 | 5.00e+00 | 4.08e+01 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 8.67e+03 | 3.39e+04
    |   3.23e-01 | 3.23e-01 |   1.11e+01 | 1.11e+01 | 6
    .88e-01 |   8.80e-01 |   8.80e-01 |
29  | LP FWH 3           | -1.53e+08 | 1.28e+07 | 1.28e+
    07 | 1.19e+01 | 1.19e+01 | 5.00e+00 | 2.35e+01 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 2.58e+03 | 3.49e+04
    |   6.17e-02 | 6.17e-02 |   1.17e+01 | 1.17e+01 | 6
    .63e-01 |   7.94e-01 |   7.94e-01 |
30  | LP FWH 4           | -2.53e+08 | 1.60e+07 | 1.60e+
    07 | 1.58e+01 | 1.58e+01 | 5.00e+00 | 3.63e+01 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 1.22e+03 | 3.60e+04
    |   7.96e-03 | 7.96e-03 |   1.22e+01 | 1.22e+01 | 6
    .55e-01 |   8.65e-01 |   8.65e-01 |

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31 | MSR drain FWH          | -2.82e+07 | 1.68e+06 | 1.68e+
    06 | 1.68e+01 | 1.68e+01 | 4.00e+01 | 5.00e+00 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 3.40e+04 | 3.19e+04
        | 1.33e+03 | 1.33e+03 | 1.02e+01 | 1.02e+01 | 8
        .86e-01 | 6.80e-03 | 8.86e-01 |
32 | interstage heater 1    | -1.07e+08 | 3.47e+06 | 3.47e+
    06 | 3.07e+01 | 3.07e+01 | 1.62e+01 | 5.20e+01 | 1.
    62e+01 | 9.70e-01 | 9.80e-01 | 1.67e+05 | 2.22e+04
        | 2.58e+03 | 2.58e+03 | 9.01e-02 | 9.01e-02 | 8
        .66e-01 | 7.09e-01 | 8.66e-01 |
33 | interstage heater 2    | -9.89e+07 | 2.11e+06 | 2.11e+
    06 | 4.68e+01 | 4.68e+01 | 3.35e+01 | 6.32e+01 | 3.
    35e+01 | 9.70e-01 | 9.80e-01 | 1.20e+05 | 2.27e+04
        | 1.69e+03 | 1.69e+03 | 1.03e-01 | 1.03e-01 | 8
        .51e-01 | 5.12e-01 | 8.51e-01 |
34 | reheater drain FWH     | -1.74e+07 | 1.57e+06 | 1.57e+
    06 | 1.11e+01 | 1.11e+01 | 2.09e+01 | 5.00e+00 | 5.
    00e+00 | 9.70e-01 | 9.70e-01 | 1.16e+05 | 1.78e+05
        | 4.52e+03 | 4.52e+03 | 5.12e+01 | 5.12e+01 | 8
        .56e-01 | 8.53e-02 | 8.56e-01 |
35 +-----+-----+-----+-----+
    ---+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+
    -----+-----+-----+-----+
36 ##### RESULTS (Valve) #####
37 +-----+-----+-----+-----+
    ---+-----+-----+-----+
38 |                               |           pr |
    dp |   zeta_d4 |   zeta |           Kv |
39 |-----+-----+-----+-----+
    ---+-----+-----+-----+-----+
40 | HP FWH drain valve 1    | 3.09e-01 | 1.34e+
    06 | 3.54e+02 | 3.54e+02 | nan      |
41 | HP FWH drain valve 2    | 7.29e-01 | 7.45e+
    05 | 6.02e+03 | 6.02e+03 | nan      |
42 | LP FWH 1 drain valve    | 4.97e-01 | 2.93e+
    05 | 8.52e+01 | 8.52e+01 | nan      |
43 | LP FWH 2 drain valve    | 3.07e-01 | 1.94e+
    05 | 6.04e+00 | 6.04e+00 | nan      |
44 | LP FWH 3 drain valve    | 4.85e-01 | 4.29e+

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44 04 | 1.27e+00 | 1.27e+00 | nan |
45 | LP FWH 4 drain valve | 1.78e-01 | 3.23e+
    04 | 7.93e-02 | 7.93e-02 | nan |
46 | MSR drain FWH drain valve | 2.63e-01 | 8.10e+
    05 | 3.44e+03 | 3.44e+03 | 2.22e+02 |
47 | interstage heater 1 drain valve | 7.18e-01 | 1.52e+
    06 | 1.56e+05 | 1.56e+05 | nan |
48 | reheater drain FWH drain valve | 7.52e-01 | 9.34e+
    05 | 5.97e+04 | 5.97e+04 | 1.63e+02 |
49 +-----+-----+-----+
    ---+-----+-----+-----+
50 ##### RESULTS (MultiStageExtractionTurbine) #####
51 +-----+-----+-----+-----+
    --+-----+-----+-----+-----+
52 | | P | pr |
    dp | eta_s1 | eta_s2 | eta_s3 | eta_s4 |
53 |-----+-----+-----+-----+
    --+-----+-----+-----+-----+
54 | HP turbine | -4.00e+08 | 7.18e-01 | 1.57e+
    06 | 8.40e-01 | 8.40e-01 | 8.40e-01 | 8.40e-01 |
55 | LP turbine stage 1 | -4.84e+08 | 2.66e-01 | 7.99e+
    05 | 8.73e-01 | 8.73e-01 | nan | nan |
56 +-----+-----+-----+-----+
    --+-----+-----+-----+-----+
57 ##### RESULTS (Turbine) #####
58 +-----+-----+-----+-----+
    --+-----+
59 | | P | pr |
    dp | eta_s |
60 |-----+-----+-----+-----+
    --+-----+
61 | LP turbine stage 2 | -1.10e+08 | 4.71e-01 | 4.55e+
    04 | 8.73e-01 |
62 | LP turbine stage 3 | -2.07e+08 | 1.73e-01 | 3.35e+
    04 | 8.73e-01 |
63 +-----+-----+-----+-----+
    --+-----+
64 ##### RESULTS (Pump) #####
65 +-----+-----+-----+-----+
    -----+-----+-----+
66 | | P | pr | dp

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66 |      eta |      eta_s |      head |
67 |-----+-----+-----+-----+
   |-----+-----+-----+-----|
68 | condenser pump | 2.83e+06 | 1.71e+02 | -1.19e+06
   | 8.04e-01 | 8.04e-01 | 1.23e+02 |
69 | feed pump      | 1.34e+07 | 6.11e+00 | -5.26e+06
   | 8.05e-01 | 8.04e-01 | 5.82e+02 |
70 +-----+-----+-----+-----+
   |-----+-----+-----+-----+
71 ##### RESULTS (CycleCloser) #####
72 +-----+-----+-----+-----+
   |--+
73 |          | mass_deviation |
   | fluid_deviation |
74 |-----+-----+-----+-----+
   |--|
75 | cycle closer |      8.95e-05 |      0.00e+
   | 00 |
76 +-----+-----+-----+-----+
   |--+
77 ##### RESULTS (Condenser) #####
78 +-----+-----+-----+-----+
   |-----+-----+-----+-----+
   |--+-----+-----+-----+-----+
   |-----+-----+-----+-----+
   |--+-----+-----+-----+-----+
79 |          |      Q |      UA |      kA
   | td_log | lmttd | ttd_u | ttd_l |
   | ttd_min | pr1 | pr2 | dp1 | dp2
   | zeta1_d4 | zeta1 | zeta2_d4 | zeta2
   | eff_hot | eff_cold | eff_max |
80 |-----+-----+-----+-----+
   |-----+-----+-----+-----+
   |--+-----+-----+-----+-----+
   |-----+-----+-----+-----+
   |--+-----+-----+-----+-----|
81 | main condenser | -2.23e+09 | 1.29e+08 | 1.29e+08
   | 1.73e+01 | 1.73e+01 | 1.20e+01 | 2.40e+01 | 1.
   | 20e+01 | 1.00e+00 | 9.80e-01 | 0.00e+00 | 2.40e+03
   | 0.00e+00 | 0.00e+00 | 1.50e-03 | 1.50e-03 |
   | 9.22e-01 | 5.00e-01 | 9.22e-01 |

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82 +-----+-----+-----+-----+
    ------+-----+-----+-----+
    ---+-----+-----+-----+-----+
    -----+-----+-----+-----+
    ---+-----+-----+
83 ##### RESULTS (SimpleHeatExchanger) #####
84 +-----+-----+-----+-----+
    -+-----+-----+
85 |           |          Q |          pr |          dp
    |  zeta_d4 |  zeta |
86 |-----+-----+-----+-----+
    -+-----+-----+
87 | steam generator 1 | 1.71e+09 | 9.70e-01 | 1.72e+05
    | 1.31e+01 | 1.31e+01 |
88 | steam generator 2 | 1.71e+09 | 9.70e-01 | 1.72e+05
    | 1.31e+01 | 1.31e+01 |
89 +-----+-----+-----+-----+
    -+-----+-----+
90 ##### RESULTS (Connection) #####
91 +-----+-----+-----+-----+
    ------+-----+-----+-----+
    -----+-----+-----+-----+
92 |
    |           |          m |          p
    |          h |          T |          x | phase |
93 |-----+-----+-----+-----+
    ------+-----+-----+-----+
    -----+-----+-----+-----+
94 | 11
    |
    |          | 4.441e+04 | 1.200e+05 | 6.309e+04
    | 2.881e+02 | 0.000e+00 | 1          |
95 | 12
    |
    |          | 4.441e+04 | 1.176e+05 | 1.133e+05
    | 3.001e+02 | 0.000e+00 | 1          |
96 | HP FWH 1:out1_HP FWH drain valve 1:in1
    |          | 5.014e+02 | 1.
    940e+06 | 9.015e+05 | 4.840e+02 | 0.000e+00 | tp
    |
97 | HP FWH 1:out2_HP FWH 2:in2

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97 | 1.
891e+03 | 6.104e+06 | 8.873e+05 | 4.805e+02 | 0.000e
+00 | 1 |
98 | HP FWH 2 shell merge:out1_HP FWH 2:in1
| 2.186e+02 | 2.
830e+06 | 1.706e+06 | 5.038e+02 | 3.937e-01 | tp
|
99 | HP FWH 2:out1_HP FWH drain valve 2:in1
| 2.186e+02 | 2.
745e+06 | 9.854e+05 | 5.021e+02 | 0.000e+00 | tp
|
100 | HP FWH 2:out2_reheater drain FWH:in2
| 1.891e+03 | 5.
921e+06 | 9.706e+05 | 4.988e+02 | 0.000e+00 | 1
|
101 | HP FWH drain valve 1:out1_LP FWH 1:in1
| 5.014e+02 | 6.
000e+05 | 9.015e+05 | 4.320e+02 | 1.108e-01 | tp
|
102 | HP FWH drain valve 2:out1_HP FWH merge:in1
| 2.186e+02 | 2.000e+06
| 9.854e+05 | 4.855e+02 | 4.069e-02 | tp |
103 | HP FWH merge:out1_HP FWH 1:in1
| 5.014e+02
| 2.000e+06 | 1.909e+06 | 4.855e+02 | 5.296e-01 |
tp |
104 | HP turbine:out1_interstage heater 2:in1
| 5.980e+01 | 4.000e
+06 | 2.732e+06 | 5.235e+02 | 9.598e-01 | tp |
105 | HP turbine:out2_HP FWH 2 shell merge:in1
| 9.250e+01 | 2.830e+
06 | 2.677e+06 | 5.038e+02 | 9.305e-01 | tp |
106 | HP turbine:out3_HP FWH merge:in2
| 2.828e+02
| 2.000e+06 | 2.624e+06 | 4.855e+02 | 9.077e-01 |
tp |
107 | HP turbine:out4_moisture separator:in1
| 1.389e+03 | 1.
133e+06 | 2.540e+06 | 4.585e+02 | 8.788e-01 | tp
|
108 | LP FWH 1 drain valve:out1_LP FWH 2 shell merge:in1

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108 | 5.014e+02 | 2.890e+05 | 6.
    652e+05 | 4.054e+02 | 5.040e-02 | tp |
109 | LP FWH 1:out1_LP FWH 1 drain valve:in1
    | 5.014e+02 | 5.
    820e+05 | 6.652e+05 | 4.308e+02 | 0.000e+00 | tp
    |
110 | LP FWH 1:out2_MSR drain FWH:in2
    | 1.891e+03
    | 1.062e+06 | 5.979e+05 | 4.151e+02 | 0.000e+00 | L
    |
111 | LP FWH 2 drain valve:out1_LP FWH 3 shell merge:in1
    | 7.741e+02 | 8.600e+04 | 5.
    516e+05 | 3.686e+02 | 6.686e-02 | tp |
112 | LP FWH 2 shell merge:out1_LP FWH 2:in1
    | 7.741e+02 | 2.
    890e+05 | 9.315e+05 | 4.054e+02 | 1.733e-01 | tp
    |
113 | LP FWH 2:out1_LP FWH 2 drain valve:in1
    | 7.741e+02 | 2.
    803e+05 | 5.516e+05 | 4.044e+02 | 0.000e+00 | tp
    |
114 | LP FWH 2:out2_LP FWH 1:in2
    | 1.
    891e+03 | 1.095e+06 | 5.352e+05 | 4.004e+02 | 0.000e
    +00 | L |
115 | LP FWH 3 drain valve:out1_LP FWH 4 shell merge:in1
    | 7.894e+02 | 4.050e+04 | 3.
    965e+05 | 3.493e+02 | 3.348e-02 | tp |
116 | LP FWH 3 shell merge:out1_LP FWH 3:in1
    | 7.894e+02 | 8.
    600e+04 | 5.902e+05 | 3.686e+02 | 8.387e-02 | tp
    |
117 | LP FWH 3:out1_LP FWH 3 drain valve:in1
    | 7.894e+02 | 8.
    342e+04 | 3.965e+05 | 3.678e+02 | 0.000e+00 | tp
    |
118 | LP FWH 3:out2_LP FWH 2:in2
    | 1.
    891e+03 | 1.129e+06 | 3.797e+05 | 3.636e+02 | 0.000e
    +00 | L |
119 | LP FWH 4 drain valve:out1_condenser merge:in2

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119 | 8.787e+02 | 7.000e+03 |
    | 3.158e+05 | 3.121e+02 | 6.330e-02 | tp |
120 | LP FWH 4 shell merge:out1_LP FWH 4:in1
    | 8.787e+02 | 4.
    | 050e+04 | 6.041e+05 | 3.493e+02 | 1.231e-01 | tp
    |
121 | LP FWH 4:out1_LP FWH 4 drain valve:in1
    | 8.787e+02 | 3.
    | 928e+04 | 3.158e+05 | 3.486e+02 | 0.000e+00 | tp
    |
122 | LP FWH 4:out2_LP FWH 3:in2
    | 1.
    | 891e+03 | 1.164e+06 | 2.988e+05 | 3.443e+02 | 0.000e
    | +00 | 1 |
123 | LP stage 1 exhaust splitter:out1_LP FWH 3 shell
    | 1.534e+01 | 8.600e+04 | 2.
    | merge:in2 | 538e+06 | 3.686e+02 | 9.426e-01 | tp |
124 | LP stage 1 exhaust splitter:out2_LP turbine stage
    | 1.101e+03 | 8.600e+04 | 2.
    | 2:in1 | 538e+06 | 3.686e+02 | 9.426e-01 | tp |
125 | LP stage 2 exhaust splitter:out1_LP FWH 4 shell
    | 8.934e+01 | 4.050e+04 | 2.
    | merge:in2 | 438e+06 | 3.493e+02 | 9.145e-01 | tp |
126 | LP stage 2 exhaust splitter:out2_LP turbine stage
    | 1.012e+03 | 4.050e+04 | 2.
    | 3:in1 | 438e+06 | 3.493e+02 | 9.145e-01 | tp |
127 | LP turbine stage 1:out1_LP FWH 2 shell merge:in2
    | 1.043e+02 | 2.890e+05 | 2.
    | 716e+06 | 4.054e+02 | 9.966e-01 | tp |
128 | LP turbine stage 1:out2_LP stage 1 exhaust
    | 1.117e+03 | 8.600e+04
    | splitter:in1 | 2.538e+06 | 3.686e+02 | 9.426e-01 | tp |
129 | LP turbine stage 2:out1_LP stage 2 exhaust
    | 1.101e+03 | 4.050e+04
    | splitter:in1 | 2.438e+06 | 3.493e+02 | 9.145e-01 | tp |
130 | LP turbine stage 3:out1_condenser merge:in1
    | 1.012e+03 | 7.000e+03
    | | 2.234e+06 | 3.121e+02 | 8.598e-01 | tp |
131 | MSR drain FWH drain valve:out1_LP FWH 2 shell
    | 1.684e+02 | 2.890e+05 | 6
    | merge:in3 | .194e+05 | 4.054e+02 | 2.925e-02 | tp |

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132 | MSR drain FWH:out1_MSR drain FWH drain valve:in1
      | 1.684e+02 | 1.099e+06 | 6.
      194e+05 | 4.201e+02 | 0.000e+00 | l      |
133 | MSR drain FWH:out2_feed pump:in1
      | 1.891e+03
      | 1.030e+06 | 6.128e+05 | 4.186e+02 | 0.000e+00 | l
      |
134 | condenser merge:out1_main condenser:in1
      | 1.891e+03 | 7.000e
      +03 | 1.342e+06 | 3.121e+02 | 4.896e-01 | tp      |
135 | condenser pump:out1_LP FWH 4:in2
      | 1.891e+03
      | 1.200e+06 | 1.648e+05 | 3.123e+02 | 0.000e+00 | l
      |
136 | cycle closer:out1_main steam splitter:in1
      | 1.891e+03 | 5.571e+
      06 | 2.786e+06 | 5.439e+02 | 9.979e-01 | tp      |
137 | feed pump:out1_HP FWH 1:in2
      | 1.891e
      +03 | 6.293e+06 | 6.199e+05 | 4.194e+02 | 0.000e+00
      | l      |
138 | feedwater splitter:out1_steam generator 1:in1
      | 9.453e+02 | 5.743e+06 |
      9.798e+05 | 5.008e+02 | 0.000e+00 | l      |
139 | feedwater splitter:out2_steam generator 2:in1
      | 9.453e+02 | 5.743e+06 |
      9.798e+05 | 5.008e+02 | 0.000e+00 | l      |
140 | interstage heater 1 drain valve:out1_interstage
      heater drain merge:in1 | 6.634e+01 | 3.880e+06 | 1.
      179e+06 | 5.217e+02 | 5.842e-02 | tp      |
141 | interstage heater 1:out1_interstage heater 1 drain
      valve:in1 | 6.634e+01 | 5.404e+06 | 1.
      179e+06 | 5.420e+02 | 0.000e+00 | tp      |
142 | interstage heater 1:out2_LP turbine stage 1:in1
      | 1.221e+03 | 1.088e+06 | 2.
      950e+06 | 5.277e+02 | 1.000e+00 | g      |
143 | interstage heater 2:out1_interstage heater drain
      merge:in2 | 5.980e+01 | 3.880e+06 | 1.
      079e+06 | 5.217e+02 | 0.000e+00 | tp      |
144 | interstage heater 2:out2_interstage heater 1:in2
      | 1.221e+03 | 1.110e+06 | 2.

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144 863e+06 | 4.900e+02 | 1.000e+00 | g      |
145 | interstage heater drain merge:out1_reheater drain
    FWH:in1          | 1.261e+02 | 3.880e+06 | 1.
    132e+06 | 5.217e+02 | 3.072e-02 | tp      |
146 | main condenser:out1_condenser pump:in1
    | 1.891e+03 | 7.
    000e+03 | 1.634e+05 | 3.121e+02 | 0.000e+00 | 1
    |
147 | main steam merge:out1_cycle closer:in1
    | 1.891e+03 | 5.
    571e+06 | 2.786e+06 | 5.439e+02 | 9.979e-01 | tp
    |
148 | main steam splitter:out1_HP turbine:in1
    | 1.824e+03 | 5.571e
    +06 | 2.786e+06 | 5.439e+02 | 9.979e-01 | tp      |
149 | main steam splitter:out2_interstage heater 1:in1
    | 6.634e+01 | 5.571e+06 | 2.
    786e+06 | 5.439e+02 | 9.979e-01 | tp      |
150 | moisture separator:out1_MSR drain FWH:in1
    | 1.684e+02 | 1.133e+
    06 | 7.869e+05 | 4.585e+02 | 1.956e-08 | tp      |
151 | moisture separator:out2_interstage heater 2:in2
    | 1.221e+03 | 1.133e+06 | 2.
    782e+06 | 4.585e+02 | 1.000e+00 | g      |
152 | reheater drain FWH drain valve:out1_HP FWH 2 shell
    merge:in2        | 1.261e+02 | 2.830e+06 | 9.
    934e+05 | 5.038e+02 | 9.469e-05 | tp      |
153 | reheater drain FWH:out1_reheater drain FWH drain
    valve:in1        | 1.261e+02 | 3.764e+06 | 9.
    934e+05 | 5.038e+02 | 0.000e+00 | 1      |
154 | reheater drain FWH:out2_feedwater splitter:in1
    | 1.891e+03 | 5.743e+06 | 9
    .798e+05 | 5.008e+02 | 0.000e+00 | 1      |
155 | steam generator 1:out1_main steam merge:in1
    | 9.453e+02 | 5.571e+06
    | 2.786e+06 | 5.439e+02 | 9.979e-01 | tp      |
156 | steam generator 2:out1_main steam merge:in2
    | 9.453e+02 | 5.571e+06
    | 2.786e+06 | 5.439e+02 | 9.979e-01 | tp      |
157 +-----+-----+-----+-----+-----+-----+-----+-----+
    -----+-----+-----+-----+-----+-----+-----+-----+

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157 -----+-----+-----+-----+
158 main steam flow      : 1890.6 kg/s
159 SG 1 duty           : 1707.0 MW
160 SG 2 duty           : 1707.0 MW
161 total heat input    : 3414.0 MW
162 gross turbine power : 1201.0 MW
163 pump power          : 16.2 MW
164 net power           : 1184.8 MW
165 cycle efficiency    : 34.70 %
166 RuntimeWarning: divide by zero encountered in divide
167 RuntimeWarning: invalid value encountered in divide
168 RuntimeWarning: invalid value encountered in
    multiply
169
170 Process finished with exit code 0
171
```